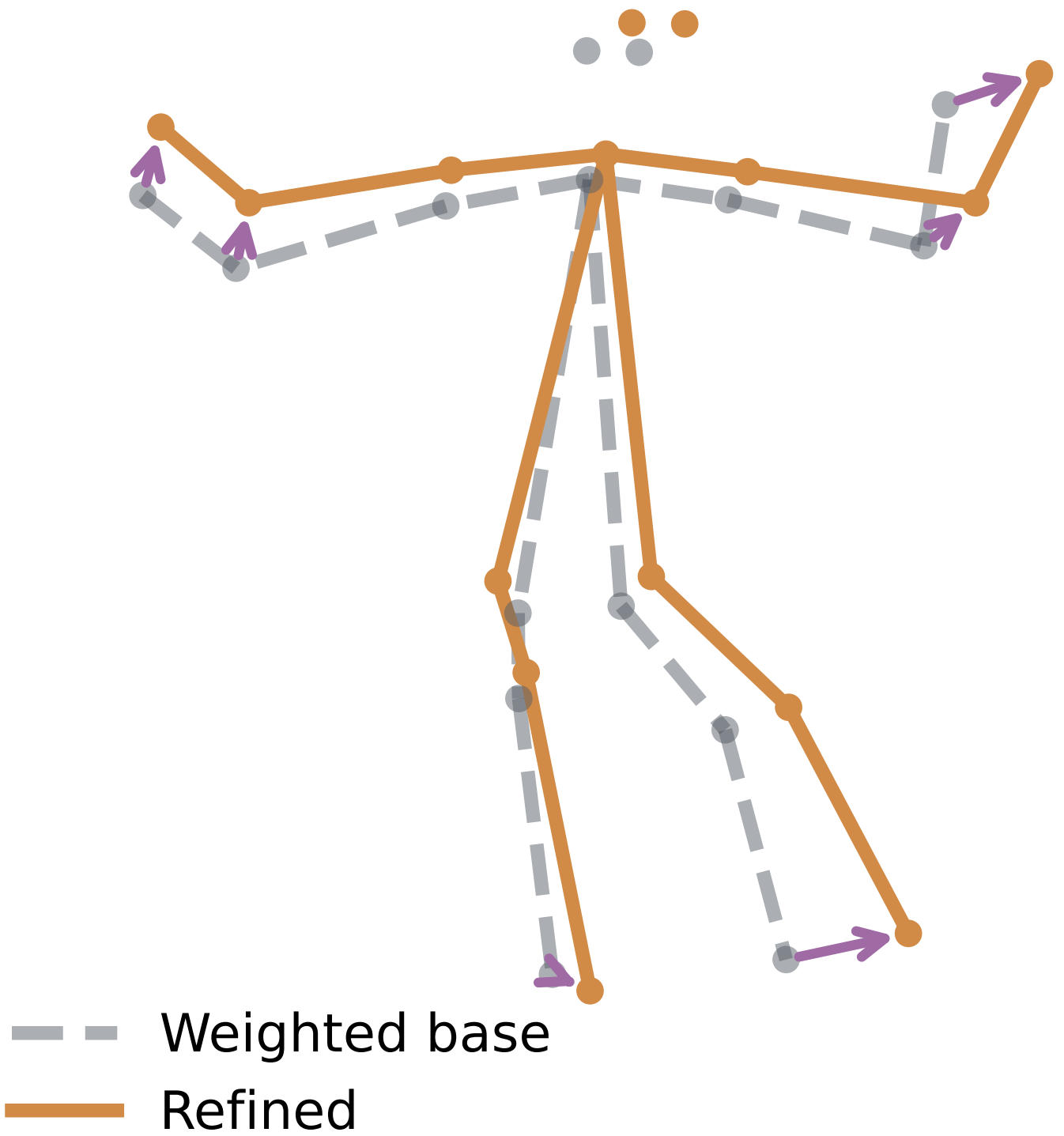Residual correction